to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promote growth by increasing the supply or availability of primary nutrients to the host plant.

23.3.1 Types of Biofertilizers

***Rhizobium*:** *Rhizobium* is a soil bacterium that colonize the roots of leguminous plants to form root nodules. The bacteria fix atmospheric nitrogen and convert them to ammonia.



Figure 23.7 *Rhizobium* biofertilizer

***Azospirillum*:** *Azospirillum* bacteria has the ability to use atmospheric nitrogen and transport this nutrient to the crop plants. It is inoculated on maize, barley, oats and sorghum crops. It increases productivity of cereals by 5 - 20%, of millets by 30% and fodder by over 50%.



Figure 23.8 *Azospirillum* biofertilizer

***Azotobacter*:** Application of *Azotobacter* bacteria has been found to increase yield of wheat, rice, maize and sorghum. Apart from nitrogen fixation, these organisms are capable of producing antifungal and antibacterial compounds.

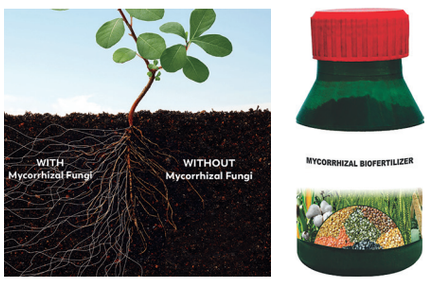


Figure 23.9 *Mycorrhizae* biofertilizer

***Mycorrhizae*:** These fungi have symbiotic association with the roots of vascular plants. They increase the uptake of phosphorus. e.g. Citrus, Papaya.

***Azolla*:** *Azolla* is a free floating, aquatic fern found on water surfaces having a cyanobacterial symbiotic association with *Anabaena*. It is a live floating nitrogen factory using energy from photosynthesis to fix atmospheric nitrogen.

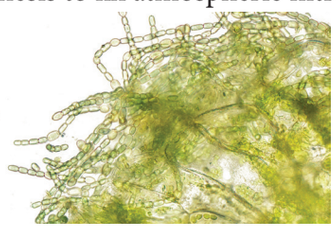


Figure 23.10 *Azolla* biofertilizers

Info bits Biofertilizer Scheme

Tamil Nadu Government has recently launched '**Biofertiliser Scheme**'. It is aimed at better management of natural farming and helps to boost and maintain soil fertility.

23.4 Medicinal Plants

The history of medicinal plants is as old as the history of human beings. Most medicines are obtained either directly or indirectly from plants. All the major system of medicines such as Ayurveda, Yoga, Unani, Siddha, Homeopathy (AYUSH) use drugs obtained from plants and animals. These drugs from medicinal plants are called secondary metabolites. Plants produce primary metabolites for their own living e.g. carbohydrates, amino acids etc., and secondary metabolites for protection, competition and species interaction. e.g. alkaloids, terpenoids, flavonoids etc. Phytochemistry is the study of phytochemicals which are chemical substances derived from various parts of the plant. Few plant derived drugs are described in (Table 23.1).

Activity 2

Collect at least five medicinal plants from your locality. Identify the plant and try to find out its medicinal value.

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